Direct Numerical Simulations of Mixed-Phase Cloud Glaciation Caused by Turbulent Mixing

Abdullah Al Muti Sharfuddin[1], Foluso Ladeinde[2], Yangang Liu[3], Fan Yang[4], Mohammad Atif[5], Satoshi Endo[6], and Meifeng Lin[7]

[1,2]Department of Mechanical Engineering, 113 Light Engineering Building, Stony Brook University, 100 Nicolls Road, Stony Brook, New York, 11794-2300, USA.

[3,4,6]Environmental Science and Technologies Department, Brookhaven National Laboratory, Rutherford Dr, Upton, New York, 11973-5000, USA.

[5,7]Computing and Data Sciences Directorate, Brookhaven National Laboratory, 2 Center St, Upton, New York, 11973-5000, USA.

Corresponding author: Foluso Ladeinde (foluso.ladeinde@stonybrook.edu)

Key Points:

- The increase of glaciation fraction follows a Sigmoidal profile. Three regimes can be distinguished during glaciation.
- A higher level of flow turbulence increases the evaporation of cloud droplets and thereby glaciation.
- A hexagonal column ice crystal grows faster than a hexagonal plate crystal, which in turn grows faster than an equivalent-volume sphere.

Abstract

The glaciation of a mixed-phase cloud is investigated at the microscale by use of direct numerical simulations (DNS) and Lagrangian particle tracking. All scales of turbulence are resolved while cloud droplets and ice crystals are tracked individually. The dry air is entrained into the cloud, and subsequent turbulent mixing is simulated. The effects of turbulence intensity, ice number density, and ice crystal geometry on glaciation are studied. It is found that high turbulence promotes cloud droplet evaporation. While non-spherical geometry promotes ice growth and glaciation, turbulence enhances glaciation by a bigger margin. Also, a higher ice number density enhances glaciation. For the simulated parameters, the increase of glaciation fraction follows a Sigmoidal profile. Three regimes can be distinguished during glaciation, governed by the competing rates of evaporation and deposition. Unlike cloud droplets, which evaporate at a nearly uniform rate, the growth rates of individual ice crystals differ substantially.

1 Introduction

Mixed-phase clouds are found at temperatures between $-40°$ to $0°$ C (Korolev et al., 2017). In this temperature range, the saturation vapor pressure and supersaturation are higher with respect to water than they are with respect to ice. Hence, the air can be supersaturated with respect to ice but subsaturated with respect to water. That will cause ice crystals to grow but cloud droplets to shrink in a mixed-phase cloud. The increase of ice mass at the expense of liquid mass is called the Wegener-Bergeron-Findeisen (WBF) process (Storelvmo & Tan, 2015) and is commonly known as glaciation. Mixed-phase clouds can fully glaciate or persist in a mixed-phase state (Morrison et al., 2012). The glaciation or persistence of mixed-phase clouds is a source of uncertainty in climate models, and it needs to be investigated at the microscale.

Mixed-phase cloud was studied analytically by Korolev (2007) who suggested that cloud droplets and ice crystals grow simultaneously if supersaturation is positive over ice and water. In that scenario, both cloud droplets and ice crystals compete for water vapor. But during the WBF process, cloud droplets feed the depositional growth of ice crystals by providing water vapor through evaporation. The analytical investigation by Pinsky et al. (2014) suggested that the mass of mixed-phase cloud increase or decrease during glaciation depending on the mass and number density of cloud droplets and ice crystals. The microphysics of mixed-phase clouds have been examined numerically. Wang et al. (2024) studied the glaciation process using a large eddy simulation (LES) model and reported that the ice mass fraction does not reach 100 % as cloud droplets are continuously brought to the mixed-phase cloud interior by turbulence. Yang et al. (2015) applied LES to examine the growth of ice crystals in a mixed-phase cloud and found that a fraction of ice crystals gets trapped in turbulent eddies and grow or sublimate faster depending on the local supersaturation over ice. Based on their LES results, Ovchinnikov et al. (2014) suggested that ice number density affects the growth of a mixed-phase cloud and so does the non-uniform distribution of cloud droplets and ice crystals. Chen et al. (2023) investigated the evolution of mixed-phase clouds using direct numerical simulations (DNS) and found that the size distributions of cloud droplets are more sensitive to turbulence than that of ice crystals. Moreover, the initial liquid mass must be large enough for the mixed-phase condition to persist.

The geometry of ice crystals is important. Using a theoretical approach, Chen & Lamb (1994) showed that the aspect ratio of ice crystal changes as the ice mass grows with time. Bailey & Hallett (2004) experimentally demonstrated that the complexity of ice crystal shape increases if the supersaturation with respect to ice is higher. Sheridan et al. (2009) suggested that ice growth is isotropic if ice crystals are larger but is anisotropic for smaller ice crystals. Sulia & Harrington

(2011) found that glaciation can be under-predicted by nearly an order of magnitude if ice crystals are assumed to be spherical instead of the precise shape. Hence, further investigation is needed to determine the effects of ice crystal shape on glaciation.

In the present study, we implement DNS with Lagrangian tracking of cloud droplets and ice crystals to study the glaciation of mixed-phase clouds. The objective is to better understand the interaction between turbulence, cloud droplets, and ice crystals during glaciation. We modify our previous particle-based DNS model (Sharfuddin et al., 2026) to solve the ice phase equations. There are two novel aspects of this study. First, we simulate the entrainment of dry air into the mixed-phase cloud and subsequent glaciation with variation of turbulence intensity and ice number density. Second, we study the role of three ice crystal geometries: hexagonal column, hexagonal plate, and sphere, on the glaciation process.

2 Governing Equations

The turbulent flow of air is governed by the incompressible Navier-Stokes equations:

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u}.\nabla)\boldsymbol{u} = -\frac{1}{\rho_a}\nabla p + \nu\nabla^2\boldsymbol{u} + \boldsymbol{F}_b + \boldsymbol{F}_c\,, \tag{1}$$

$$\nabla.\boldsymbol{u} = 0\,. \tag{2}$$

Here, $\boldsymbol{u}$ is the instantaneous air velocity vector, ν is the kinematic viscosity of air, p is the instantaneous air pressure, and ρ_a is the air density. $\boldsymbol{F}_c$ is an external force that maintains flow turbulence and is constructed in the Fourier space following the procedure of Eswaran and Pope (1988). Although the buoyancy force $\boldsymbol{F}_b$ is included, it is negligible in our microscale model. The definitions of $\boldsymbol{F}_c$ and $\boldsymbol{F}_b$ can be found in Sharfuddin et al. (2026). The conservation equations for the temperature (T) and water vapor mixing ratio (q_v) are:

$$\frac{\partial T}{\partial t} + (\boldsymbol{u}.\nabla)T = \frac{L_w}{c_p}C_w + \frac{L_{ice}}{c_p}C_{ice} + \mu_T\nabla^2 T\,, \tag{3}$$

$$\frac{\partial q_v}{\partial t} + (\boldsymbol{u}.\nabla)q_v = -(C_w + C_{ice}) + \mu_v\nabla^2 q_v\,. \tag{4}$$

Above, L_w and L_{ice} are specific latent heats of condensation and deposition, respectively. C_w is the time rate of exchange between water vapor and liquid water while C_{ice} is the same between water vapor and solid ice. c_p is the specific heat of air and μ_T and μ_v are the molecular diffusivities of temperature and vapor, respectively. The environmental supersaturation (S_e) depends on the T and q_v, and is expressed as:

$$S_e = \frac{q_v}{q_{v,s}} - 1 = \frac{q_v}{621.97\frac{p_{sat}}{p - p_{sat}}} - 1\,, \tag{5}$$

where $q_{v,s}$ is the saturation vapor mixing ratio and p_{sat} is the saturation vapor pressure which can be calculated with respect to water and ice (Huang, 2018) as:

$$p_{sat} = \begin{cases} p_{sat,w} = 611.2 \times \exp\left(\dfrac{17.67(T - 273.15)}{T - 29.65}\right) \\ p_{sat,ice} = 611.2 \times \exp\left(\dfrac{22.46(T - 273.15)}{T - 0.53}\right) \end{cases}. \tag{6}$$

The cloud droplets and ice crystals are tracked individually in Lagrangian fashion. The equations for the Lagrangian position ($\boldsymbol{X}$) and velocity ($\boldsymbol{V}$) vectors are written as:

$$\frac{d\boldsymbol{X}_i(t)}{dt} = \boldsymbol{V}_i(t)\,, \quad \frac{d\boldsymbol{V}_i(t)}{dt} = \frac{1}{\left(\tau_p\right)_i}\left[\boldsymbol{u}_i(\boldsymbol{X},t) - \boldsymbol{V}_i(t)\right] + \boldsymbol{g}\,, \quad \left(\tau_p\right)_i = \frac{2\rho_i (r_i)^2}{9\rho_a \nu}\,. \tag{7}$$

where the subscript 'i' denotes the i-th particle (cloud droplet or ice crystal). $\boldsymbol{u}_i$ is the instantaneous air velocity vector interpolated at the position of particle i, $\boldsymbol{g}$ is the gravity vector, τ_p is the particle response time, and ρ_i is the density of i-th particle. Conveniently, S_e and particle radius (r) can be written as:

$$S_e = \begin{cases} S_{e,w} & if \quad p_{sat} = p_{sat,w} \\ S_{e,ice} & if \quad p_{sat} = p_{sat,ice} \end{cases}, \quad r = \begin{cases} r_w & if \quad S_e = S_{e,w} \\ r_{ice} & if \quad S_e = S_{e,ice} \end{cases}. \tag{8}$$

The growth equations for cloud droplets and ice crystals are written as:

$$\frac{d\left(r_w(t)\right)_i}{dt} = \left(\frac{G_w}{r_w(t)} S_{e,w}(\boldsymbol{X},t)\right)_i\,, \quad \frac{d\left(r_{ice}(t)\right)_i}{dt} = \left(\frac{G_{ice}}{r_{ice}(t)} S_{e,ice}(\boldsymbol{X},t)\right)_i\,. \tag{9}$$

The growth factors G_w and G_{ice} are expressed as:

$$G_w = \frac{1}{\frac{L_w \rho_w}{k_T T}\left(\frac{L_w}{R_v T} - 1\right) + \frac{\rho_w R_v T}{\mu_v p_{sat,w}}}\,, \quad G_{ice} = f\left[\frac{1}{\frac{L_{ice}\rho_{ice}}{k_T T}\left(\frac{L_{ice}}{R_v T} - 1\right) + \frac{\rho_{ice} R_v T}{\mu_v p_{sat,ice}}}\right]. \tag{10}$$

where k_T is the thermal conductivity in air, R_v is the universal gas constant divided by the molecular weight of water, and f is the ice crystal shape factor. The f is unity for spherical ice crystals. For hexagonal ice crystals, the f is found from: (Westbrook et al., 2008)

$$f = 0.58(1 + 0.95\mathrm{A}^{0.75})(b/r_{ice})\,, \quad \mathrm{A} = \frac{h}{2b}\,. \tag{11}$$

Above, b is the side length and h is the height of a regular hexagonal prism, and A is the aspect ratio. The shape is hexagonal column if $\mathrm{A} > 1$ and hexagonal plate if $\mathrm{A} < 1$. r_{ice} is the equivalent radius of hexagonal ice crystals and is related to b and h by:

$$\frac{4}{3}\pi r_{ice}^3 = \frac{3\sqrt{3}}{2} b^2 h\,, \quad r_{ice} = 0.8527(b^2 h)^{1/3}\,, \quad b = 0.93\left(\frac{r_{ice}}{\mathrm{A}^{1/3}}\right). \tag{12}$$

Our model neglects ice nucleation, accretion, aggregation, and ventilation. It focuses on diffusion-limited ice growth only. With the foregoing, C_w and C_{ice} are determined as:

$$C_w(\boldsymbol{x},t) = \frac{4\pi\rho_w}{\rho_a a^3}\sum_{i=1}^{n_w}\left(G_w r_w(t) S_{e,w}(\boldsymbol{X},t)\right)_i\,, \tag{13}$$

$$C_{ice}(\boldsymbol{x},t) = \frac{4\pi\rho_{ice}}{\rho_a a^3}\sum_{i=1}^{n_{ice}}\left(G_{ice} r_{ice}(t) S_{e,ice}(\boldsymbol{X},t)\right)_i\,. \tag{14}$$

where ρ_w and ρ_{ice} are the densities of liquid water and solid ice, respectively. n_w is the number of cloud droplets within a grid cell having a volume of a^3 and n_{ice} is the same for ice crystals.

3 Numerical Details

Assuming homogeneous and isotropic turbulence (HIT), we generate two turbulent kinetic energy (TKE) dissipation rates by varying the band of forcing wavenumbers (Sharfuddin et al., 2026). The two levels are denoted as 'low' (L) and 'high' (H) and correspond to the Kolmogorov length scales of 0.0018 m and 0.00098 m. Also, the wavenumber bands are: $1 \leq k_f/k_{min} \leq 4.12$

and $1 \le k_f/k_{min} \le 16.18$, respectively, where $k_{min} = 2\pi/L = 12.27$ [1/m] and k_f is the forcing wavenumber. $L = 0.512\ m$ is the domain's side length.

The physical model is that of a cubic domain inside which turbulence, cloud droplets, and ice crystals interact. The domain volume is 0.512^3 m^3, with triply periodic boundary conditions. At the initial time, the domain is equally divided into cloudy and clear-air regions. Cloud droplets and ice crystals are randomly placed in the cloudy region. A slab-like configuration is assumed so that the initial temperature and water vapor mixing ratio change sharply at the cloud/clear-air interface. This setup allows simulation of entrainment and mixing (Gao et al., 2018). The initial vapor mixing ratio is specified as:

$$q_v(z, t = 0) = \begin{cases} q_v^{max}, & d - L \times 0.5 \times 0.5 \le z < d + L \times 0.5 \times 0.5 \\ q_{v,e}, & elsewhere \end{cases}. \quad (15)$$

Above, $q_v^{max} = 3.12\ g/kg$, $q_{v,e} = 2.50\ g/kg$, and $d \equiv L/2$ is the width of the cloud slab. q_v^{max} corresponds to the cloudy region and $q_{v,e}$ represents the dry (subsaturated) region. The temperature field is initialized as:

$$T(t = 0) = \langle T(t = 0)\rangle - 0.608\langle T(t = 0)\rangle(q_v(z, t = 0) - \langle q_v(t = 0)\rangle)\,. \quad (16)$$

We simulate five cases: L-S-L, L-C-L, L-P-L, H-S-L, and H-S-H. The details are provided in Table 1 along with the values of relevant parameters. The ice crystal shape is hexagonal column for Case L-C-L, hexagonal plate for Case L-P-L, and spherical otherwise. The initial volume of an ice crystal is approximately 4.18 μm^3 for all cases. For a hexagonal column, the initial values of f, b, and h are 1.148, 0.685 μm, and 3.42 μm, respectively. The corresponding values for a hexagonal plate are 1.082, 1.26 μm, and 1.01 μm. The domain-average temperature is ~264 K initially. The common ice crystal shape at this temperature is hexagonal column (Bailey & Hallett, 2009) but hexagonal plate shape is possible depending on the $S_{e,ice}$. The assumption of spherical ice crystals is rather idealistic but necessary for comparison. The cloud droplet number density is 59.6 cm^{-3} and is much larger than the ice number density (Korolev et al., 2017). The number of grid points is 256^3 and the grid resolution is 2 mm for all cases. It is assumed that the aspect ratio of hexagonal ice crystals does not change with time in our microscale DNS model.

Table 1

Details of the test cases

Case	TKE dissipation rate (m^2/s^3)	Number density of ice crystals (cm^{-3})	Initial radii of each cloud droplet, ice crystal (μm)	Initial surface area of an ice crystal (μm^2)	Aspect ratio of ice crystal
L-S-L	0.00068	0.011	8, 1	12.566	N/A
L-C-L	0.00068	0.011	8, 1	16.494	2.5
L-P-L	0.00068	0.011	8, 1	15.885	0.4
H-S-L	0.00816	0.011	8, 1	12.566	N/A
H-S-H	0.00816	0.055	8, 1	12.566	N/A

Case	Description
L-S-L	low (L) turbulence, spherical (S) ice crystals, low (L) ice number density
L-C-L	low (L) turbulence, hexagonal column (C) ice crystals, low (L) ice number density

L-P-L	low (L) turbulence, hexagonal plate (P) ice crystals, low (L) ice number density
H-S-L	high (H) turbulence, spherical (S) ice crystals, low (L) ice number density
H-S-H	high (H) turbulence, spherical (S) ice crystals, high (H) ice number density

Parameters and their values

Symbol	Value	Unit	Symbol	Value	Unit
L_w	2.53×10^6	J/kg	$\mu_T = \mu_v$	2.9×10^{-5}	m^2/s
L_{ice}	2.84×10^6	J/kg	c_p	1005.0	J/kg/K
ρ_w	1000	kg/m^3	ρ_{ice}	917	kg/m^3
$p(t=0)$	64455	N/m^2	$\langle T(t=0) \rangle$	264.38	K
$p_{sat,w}$	315.73	N/m^2	$p_{sat,ice}$	289.60	N/m^2
ρ_a	0.85	kg/m^3	ν	1.96×10^{-5}	m^2/s
R_v	461.5	J/kg/K	k_T	0.0238	W/m/K

4 Results and Discussion

The mean supersaturation is negative with respect to water but positive with respect to ice at the initial time. Hence, ice crystals grow but cloud droplets evaporate, and the mixed-phase cloud glaciates. The effects of flow turbulence, ice number density, and ice crystal geometry on the thermodynamics and microphysics of glaciation are discussed.

The temporal evolution of the mean supersaturation fields is shown in Figure 1(a). Liquid water is converted into water vapor during the evaporation of cloud droplets and ice crystals grow by consuming that water vapor. Since the number density and mass of cloud droplets exceed that of ice crystals in mixed-phase clouds, more water vapor is produced by evaporation than consumed by ice crystals. Corresponding to the net increase of water vapor, the $\langle S_{e,w} \rangle$ and $\langle S_{e,ice} \rangle$ increase over time as shown in Figure 1(a). We see that the increase is slightly faster when the turbulence level is high (Cases H-S-L and H-S-H). In contrast, the DNS study of Chen et al. (2023) found no impact of flow turbulence on the mean supersaturations because it did not account for entrainment and mixing like our model does. It is evident that the mean supersaturation profiles are identical for Cases L-S-L, L-P-L, and L-C-L. Thus, the ice crystal geometry has no influence on the $\langle S_{e,w} \rangle$ and $\langle S_{e,ice} \rangle$. A higher ice number density (Case H-S-H) means larger total ice crystal surface area. This allows more water vapor molecules to deposit on ice crystal surfaces, reducing the amount of water vapor and thus $\langle S_{e,w} \rangle$ and $\langle S_{e,ice} \rangle$ at late times (Figure 1(a)).

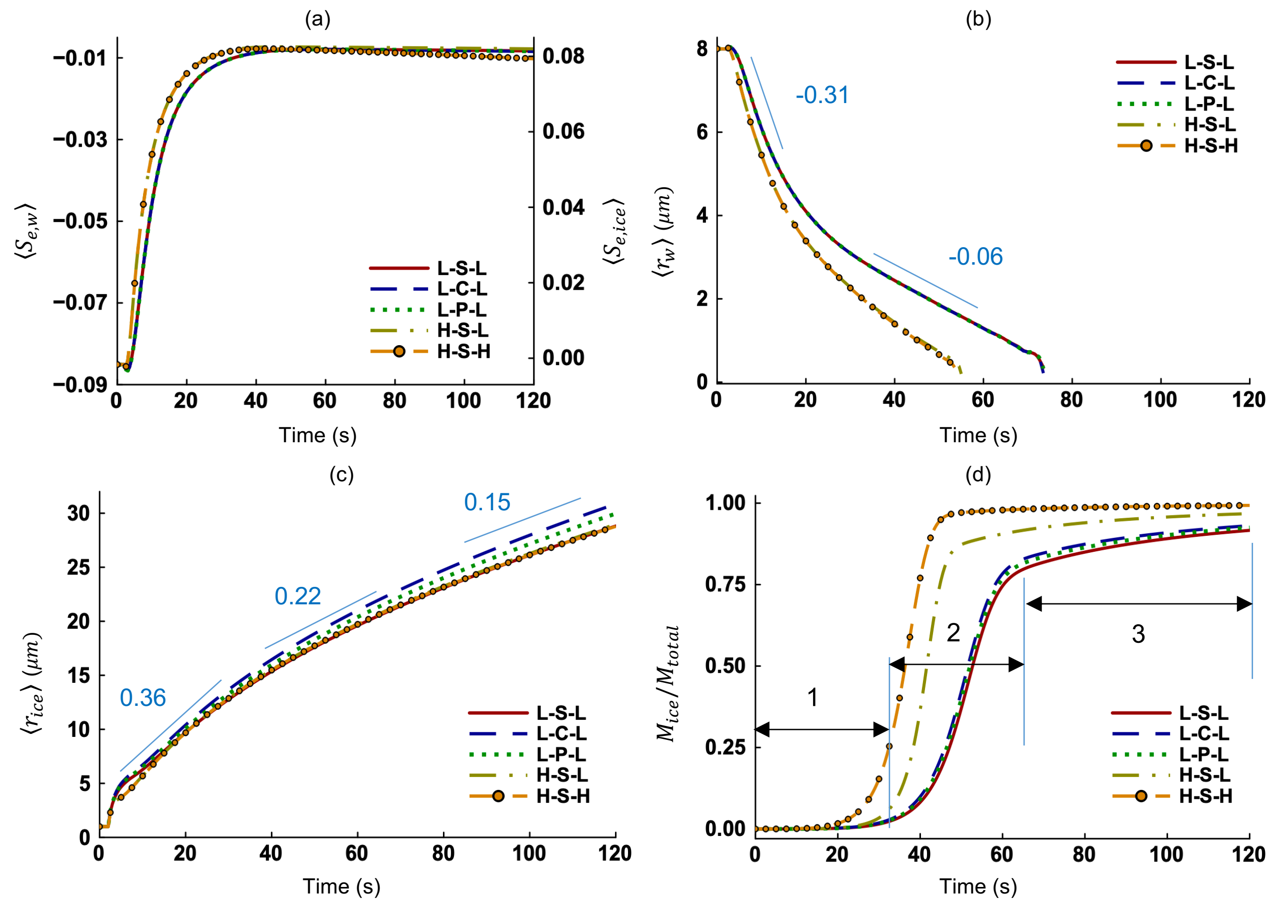
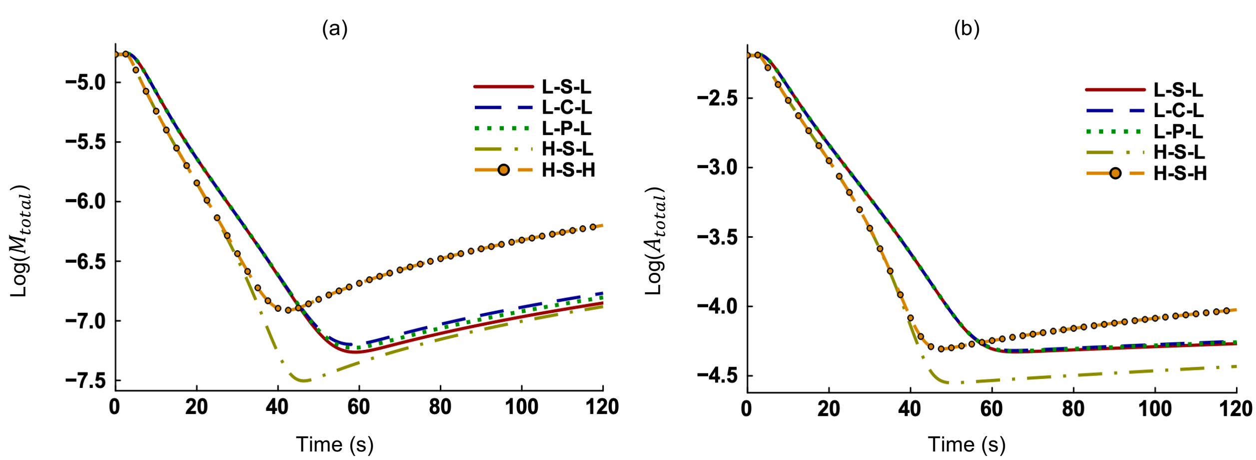


Figure 1. Time evolution of the (a) mean environmental supersaturations with respect to water ($\langle S_{e,w} \rangle$) and ice ($\langle S_{e,ice} \rangle$), (b) mean radius of cloud droplets ($\langle r_w \rangle$), (c) mean radius of ice crystals ($\langle r_{ice} \rangle$), and (d) ice mass (or glaciation) fraction. Three regimes can be identified in Figure 1(d) for all cases. We draw markers on one curve (Case L-S-L) for legibility. The plots for Cases H-S-L and H-S-H overlap in Figures 1(b) and 1(c). Also, the plots for Cases L-S-L, L-C-L, and L-P-L overlap in Figures 1(a) and 1(b).

The mean cloud droplet radius ($\langle r_w \rangle$) decreases faster (Figure 1(b)) because evaporation is stronger if the turbulence intensity is high (Cases H-S-L and H-S-H). Flow turbulence accelerates mixing and homogenization, leading to a faster evaporation of cloud droplets (Hoffmann, 2020). Evaporation is quicker between 0 s to 20 s for all five cases. Evaporation moderates after 20 s and is completed by 55 s for high turbulence cases (H-S-L and H-S-H) and by 75 s for low turbulence cases (L-S-L, L-P-L, and L-C-L). As an example, the slope of evaporative decay is -0.31 for Case L-C-L at early times and -0.06 afterwards. Evaporation is faster at early times as supersaturation is highly negative (Figure 1(a)). The domain becomes less subsaturated with time. We also report that ice crystal geometry has no impact on cloud droplet evaporation based on the identical profiles in Figure 1(b) for Cases L-S-L, L-P-L, and L-C-L. The mean radius of ice crystals ($\langle r_{ice} \rangle$) increases during deposition for all five cases (Figure 1(c)). The rate of increase slows down with time as indicated by the slopes (0.36 > 0.22 > 0.15). This happens because particle radius is inversely proportional to its growth rate (Equation (9)). The profiles of $\langle r_{ice} \rangle$ are identical for Cases L-S-L and H-S-L, implying that turbulence has negligible impact on ice crystal deposition. The number density of ice crystals is very low in mixed-phase clouds and thus they do not need to compete for water vapor. Because localized vapor diffusion is dominant, turbulent mixing is inconsequential for ice crystal growth. But ice crystal geometry affects the evolution of $\langle r_{ice} \rangle$. We see in Figure 1(c) that hexagonal column (L-C-L) and hexagonal plate (L-P-L) ice crystals grow faster than equivalent-volume spherical ice crystals (L-S-L). A sphere has uniform curvature everywhere with no sharp edges while a hexagonal crystal has sharp edges. These edges steepen the adjacent water vapor density gradients, attracting a proportionally higher flux of water vapor (Fick's first law of diffusion). Also, hexagonal column ice crsytals grow faster than hexagonal plate ice crystals. As water vapor deposits onto ice crystal surfaces, a localized vapor depletion zone is created. The hexagonal column ($A = 2.5$) is highly prolate, and its top and bottom ends grow independently without their respective vapor depletion zones overlapping. In contrast, the oblate geometry of a hexagonal plate ($A = 0.4$) causes its basal faces to reside within the vapor depletion zones of the perimeter edges, restricting water vapor deposition at the interior.

The ratio of ice crystal mass to total cloud mass is denoted by M_{ice}/M_{total}. It indicates the fraction of the mixed-phase cloud that has been glaciated. Three regimes can be differentiated for all cases in Figure 1(d). Regime (1) is evaporation-dominated as the rate of evaporation is highest at early times. The mass of ice crystals, although growing fast, is still small compared to the total cloud mass and the glaciation fraction (M_{ice}/M_{total}) is negligible. With continuous evaporation and deposition, the contribution of M_{ice} to M_{total} increases and thus the glaciation fraction rises sharply. This period marks Regime (2), during which glaciation is enhanced by both evaporation and deposition. The glaciation fraction increases at a relatively slower rate during Regime (3), which is driven by ice crystal deposition only as all cloud droplets have fully evaporated by that time. The growth profiles in Figure 1(d) resemble a Sigmoid function: $y_a/(1 + e^{-K(t-t_0)})$, where y_a is the plateau, K is the logistic growth rate, representing the steepness of the curve during rapid glaciation phase (Regime (2)), and t_0 is the time needed for 50% glaciation. The derivation of the Sigmoid function, along with corresponding values of y_a, K, and t_0 for each case, is provided in the supporting information Text S1 and Table S1. Figure 1(d) also shows that glaciation rate is noticeably higher if the turbulence intensity is high (Cases H-S-L and H-S-H) compared to low turbulence cases (L-S-L, L-P-L, and L-C-L). With more ice crystals present, the glaciation rate is highest (H-S-H). For the same turbulence level, hexagonal column (L-C-L) and hexagonal plate (L-P-L) ice crystals enhance glaciation compared to spherical ice crystals (L-S-L). It is evident that turbulent mixing more strongly contributes to glaciation than the morphology of ice crystal does.



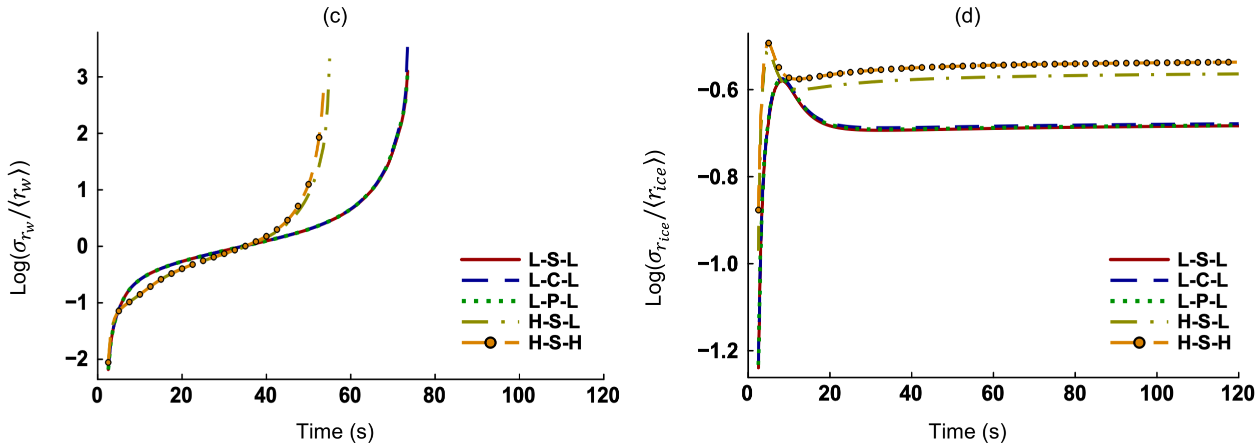
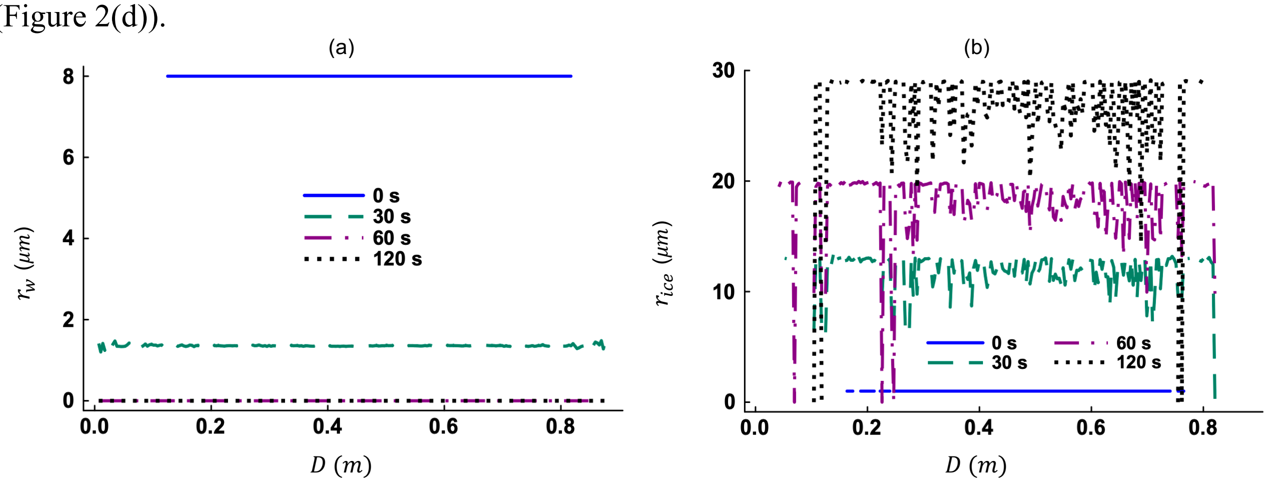


Figure 2. Time evolution of the (a) total cloud mass (M_{cloud}), (b) total cloud area (A_{cloud}), (c) relative dispersion of cloud droplet radius ($\sigma_{r_w}/\langle r_w \rangle$), and (d) relative dispersion of ice crystal radius ($\sigma_{r_{ice}}/\langle r_{ice} \rangle$). The ordinate is shown on a logarithmic scale of base 10.

The total mass and area of the mixed-phase cloud decrease first (Figures 2(a) and 2(b)), corresponding to strong evaporation, but increase afterwards since ice deposition continues and evaporation is completed. The increase of M_{total} and A_{total} is largest if both the turbulence level and ice number density are high (Case H-S-H). For low turbulence cases, the order of increase is L-C-L > L-P-L > L-S-L, consistent with Figures 1(c) and 1(d). The combination of high turbulence and low ice number density (Case H-S-L) causes least increase in the M_{total} and A_{total}. From 60 s to 120 s, cloud growth is determined by deposition and the increase of mass and area is limited by the number of ice crystals. We also analyze the relative dispersion of particle radius in liquid and ice phases. The relative dispersion of cloud droplet radii ($\sigma_{r_w}/\langle r_w \rangle$) increases with time, shoots up just before evaporation ends and vanishes when cloud droplets no longer exist (Figure 2(c)). The relative dispersion of ice crystal radii ($\sigma_{r_{ice}}/\langle r_{ice} \rangle$) increases fast initially when monodisperse ice crystals are released in a turbulent setting (Figure 2(d)). The $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ levels off afterwards, implying that the increase of standard deviation $\sigma_{r_{ice}}$ is equal to that of mean radius $\langle r_{ice} \rangle$ (Figure 1(c)). We find that higher turbulence (H-S-L and H-S-H) has an expediting effect on $\sigma_{r_{ice}}/\langle r_{ice} \rangle$ (Figure 2(d)).



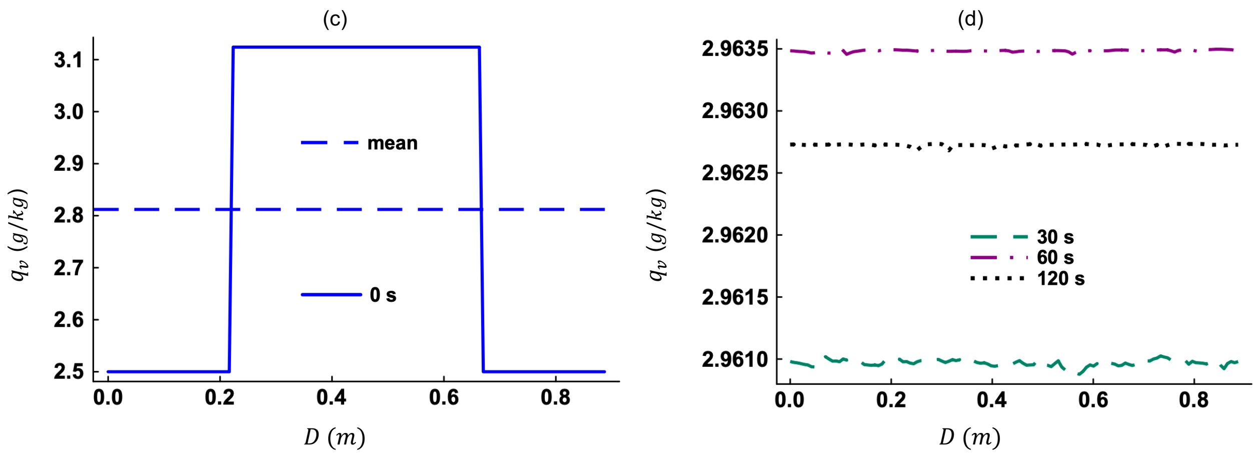
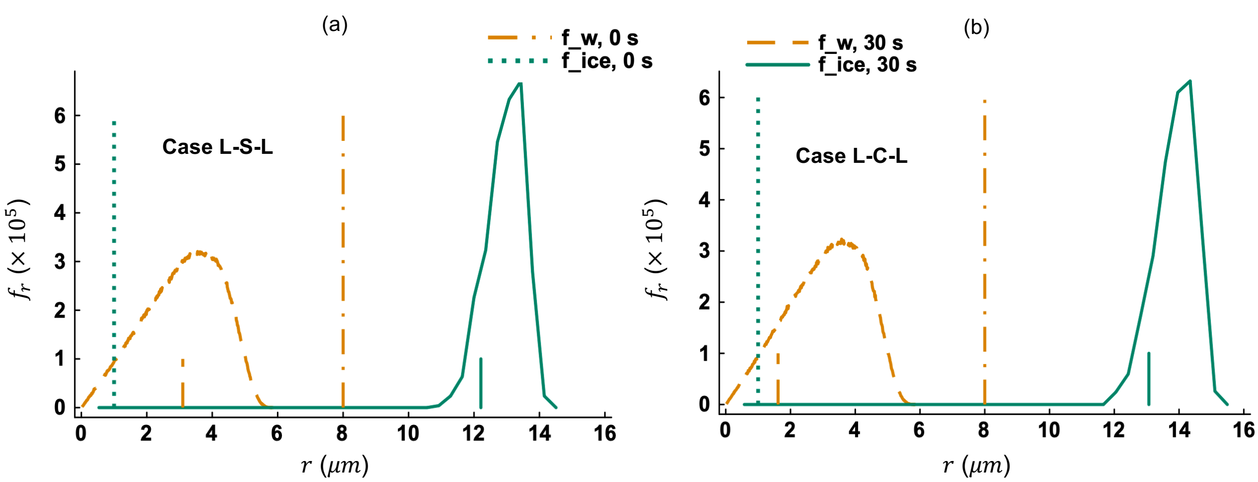


Figure 3. The spatial distributions of the (a) r_w and (b) r_{ice} at selected times. The spatial distributions of the q_v at (c) 0 s and at (d) other selected times. Note that D is the main diagonal of our cubic domain. These results represent Case H-S-L.

The spatial distributions of the r_w and r_{ice} reveal different growth histories for liquid and ice phases (Figures 3(a)-(b)). Because ice crystals are fewer in number and the domain is rich in water vapor, vapor molecules can freely deposit on ice crystal surfaces. This allows ice crystals to grow at different rates (Figure 3(b)). In contrast, cloud droplets evaporate almost at the same rate (Figure 3(a)). Evaporation is completed by 55 s for Case H-S-L (Figure 1(b)) and thus $r_w = 0$ at 60 s and 120 s in Figure 3(a). The spatial distribution of the q_v at 0 s shows that sharp interfaces are present (Figure 3(c)), which is consistent with Equation (15). From a mean value of 2.812 g/kg at 0 s, q_v increases with time as evaporation releases water vapor. However, it decreases between 60 s and 120 s (Figure 3(d)) when only deposition takes place and water vapor is converted to solid ice. We also observe that fluctuations are negligible in the q_v compared to its mean component. Evaporation and deposition act as source and sink for water vapor, respectively, strongly affecting the mean q_v. While flow turbulence does not create or destroy water vapor and merely redistributes the latter.



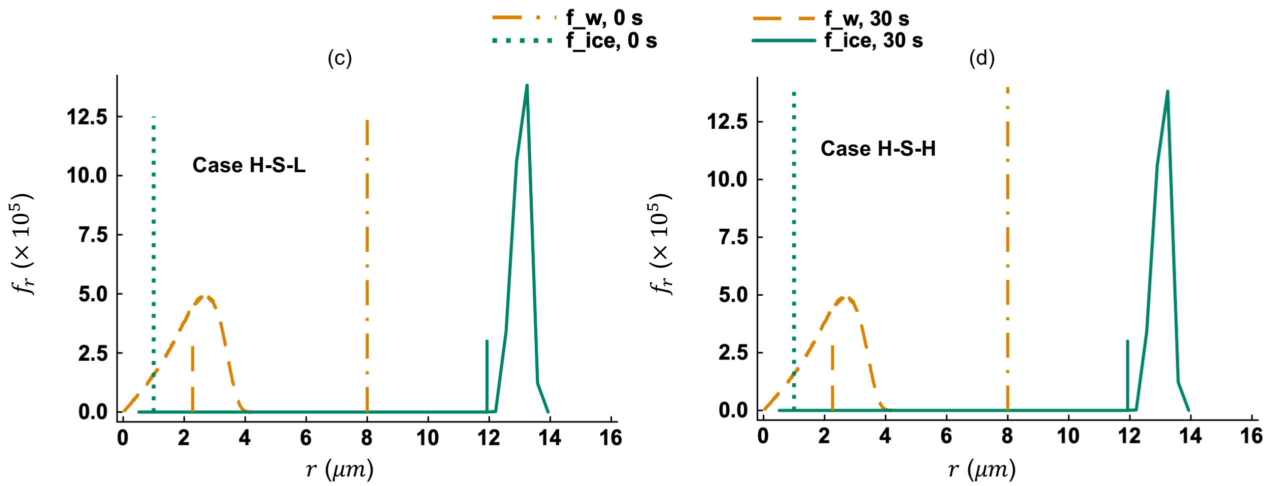


Figure 4. Probability density functions (PDFs) of particle radius (f_r) for Cases (a) L-S-L, (b) L-C-L, (c) H-S-L, and (d) H-S-H. The vertical line segments at the bottom indicate the expectation (mean) of r in the respective distributions.

Figure 4 contrasts the size distributions of cloud droplets and ice crystals in a mixed-phase cloud. The PDF of cloud droplet radii (f_{r_w}) broadens between 0 s and 30 s for all cases (Figures 4(a)-(d)) because of turbulent mixing. The f_{r_w} is narrower at 30 s for high turbulence cases (Figures 4(c)-(d)) compared to that for low turbulence cases (Figures 4(a)-(b)). The narrowing is caused by a reduction in maximum cloud droplet radius. So, strong turbulence leads to complete evaporation of more cloud droplets. Broadening is again observed in the PDF of ice crystal radii ($f_{r_{ice}}$) at 30 s compared to the monodisperse distribution at 0 s. A tiny fraction of ice crystals sublimate for all cases due to entrainment of dry air into the mixed-phase cloud volume, making the $f_{r_{ice}}$ highly skewed to the left. The mean and maximum ice crystal radii are largest when the turbulence level is low and ice crystal shape is hexagonal column (Figure 4(b)).

5 Concluding Remarks

The glaciation of a mixed-phase cloud, caused by entrainment and turbulent mixing, has been studied using a three-dimensional particle-based DNS model. Our findings clarify several microphysical aspects of the glaciation process. A high level of flow turbulence intensifies mixing between the cloud and environment and expedites glaciation. Glaciation is also enhanced if the ice number density is high. Non-spherical geometries like hexagonal column and hexagonal plate lead to higher ice growth compared to equivalent-volume spheres. This letter shows that particle-based DNS can resolve the distinct growth histories of cloud droplets and ice crystals, a capability that remains challenging for many experimental setups. Future iterations of this model will incorporate variable ice crystal aspect ratios to capture the anisotropic growth of ice crystals. More complicated ice crystal shapes like snowflake, dendrite, and rosette will be considered.

Acknowledgments

This research used the resources of the National Energy Research Scientific Computing Center (NERSC), based on the award: NERSC ASCR-ERCAP0027399. We acknowledge support from the LDRD program at Brookhaven National Laboratory, sponsored by the US Department of Energy, Office of Science, under Contract Number DE-SC0012704.

References

Bailey, M. & Hallett, J. (2004), Growth rates and habits of ice crystals between -20 and -70 C. *Journal of the Atmospheric Sciences*, *61*(5), 514-544. doi: 10.1175/1520-0469(2004)061<0514:GRAHOI>2.0.CO;2.

Bailey, M.P. & Hallett, J. (2009), A comprehensive habit diagram for atmospheric ice crystals: Confirmation from the laboratory, AIRS II, and other field studies. *Journal of the Atmospheric Sciences*, *66*(9), 2888-2899. doi: 10.1175/2009JAS2883.1.

Chen, J.P. & Lamb, D. (1994), The theoretical basis for the parameterization of ice crystal habits: Growth by vapor deposition. *Journal of Atmospheric Sciences*, *51*(9), 1206-1222. doi: 10.1175/1520-0469(1994)051<1206:TTBFTP>2.0.CO;2.

Chen, S., Xue, L., Tessendorf, S., Ikeda, K., Weeks, C., Rasmussen, R., Kunkel, M., Blestrud, D., Parkinson, S., Meadows, M. & Dawson, N. (2023), Mixed-phase direct numerical simulation: ice growth in cloud-top generating cells. *Atmospheric Chemistry and Physics*, *23*(9), 5217-5231. doi: 10.5194/acp-23-5217-2023.

Eswaran, V. & Pope, S.B. (1988), An examination of forcing in direct numerical simulations of turbulence. *Computers & Fluids*, *16*(3), pp.257-278. doi: 10.1016/0045-7930(88)90013-8.

Gao, Z., Liu, Y., Li, X. & Lu, C. (2018), Investigation of turbulent entrainment-mixing processes with a new particle-resolved direct numerical simulation model. *Journal of Geophysical Research: Atmospheres*, *123*(4), 2194-2214. doi: 10.1002/2017JD027507.

Hoffmann, F. (2020), Effects of entrainment and mixing on the Wegener–Bergeron–Findeisen process. *Journal of the Atmospheric Sciences*, *77*(6), 2279-2296. doi: 10.1175/JAS-D-19-0289.1.

Huang, J. (2018), A simple accurate formula for calculating saturation vapor pressure of water and ice. *Journal of Applied Meteorology and Climatology*, *57*(6), 1265-1272. doi: 10.1175/JAMC-D-17-0334.1.

Korolev, A. (2007), Limitations of the Wegener–Bergeron–Findeisen mechanism in the evolution of mixed-phase clouds. *Journal of the Atmospheric Sciences*, *64*(9), 3372-3375. doi: 10.1175/JAS4035.1.

Korolev, A., McFarquhar, G., Field, P.R., Franklin, C., Lawson, P., Wang, Z., Williams, E., Abel, S.J., Axisa, D., Borrmann, S. & Crosier, J. (2017). Mixed-phase clouds: Progress and challenges. *Meteorological Monographs*, *58*, pp.5-1. doi: 10.1175/AMSMONOGRAPHS-D-17-0001.1.

Morrison, H., De Boer, G., Feingold, G., Harrington, J., Shupe, M.D. & Sulia, K. (2012), Resilience of persistent Arctic mixed-phase clouds. *Nature Geoscience*, *5*(1), 11-17. doi: 10.1038/ngeo1332.

Ovchinnikov, M., Ackerman, A.S., Avramov, A., Cheng, A., Fan, J., Fridlind, A.M., Ghan, S., Harrington, J., Hoose, C., Korolev, A. & McFarquhar, G.M. (2014), Intercomparison of large-eddy simulations of Arctic mixed-phase clouds: Importance of ice size distribution assumptions. *Journal of Advances in Modeling Earth Systems*, *6*(1), pp.223-248. doi: 10.1002/2013MS000282.

Pinsky, M., Khain, A. & Korolev, A. (2014), Analytical investigation of glaciation time in mixed-phase adiabatic cloud volumes. *Journal of the Atmospheric Sciences*, *71*(11), pp.4143-4157. doi: 10.1175/JAS-D-13-0359.1.

Sharfuddin, A.A.M., Ladeinde, F., Liu, Y., Atif, M., Yang, F., Lopez-Marrero, V., Lin, M. & Zhang, T. (2026), Direct numerical simulations of activation and deactivation in turbulent atmospheric clouds. *Physics of Fluids*, *38*(1). doi: 10.1063/5.0301098.

Sheridan, L.M., Harrington, J.Y., Lamb, D. & Sulia, K. (2009), Influence of ice crystal aspect ratio on the evolution of ice size spectra during vapor depositional growth. *Journal of the atmospheric sciences*, *66*(12), 3732-3743. doi: 10.1175/2009JAS3113.1.
Storelvmo, T. & Tan, I. (2015), The Wegener-Bergeron-Findeisen process–Its discovery and vital importance for weather and climate. *Meteorologische Zeitschrift*, *24*(4), 455-461. doi: 10.1127/metz/2015/0626.
Sulia, K.J. & Harrington, J.Y. (2011), Ice aspect ratio influences on mixed-phase clouds: Impacts on phase partitioning in parcel models. *Journal of Geophysical Research: Atmospheres*, *116*(D21). doi: 10.1029/2011JD016298.
Wang, A., Krueger, S., Chen, S., Ovchinnikov, M., Cantrell, W. & Shaw, R.A. (2024), Glaciation of mixed-phase clouds: insights from bulk model and bin-microphysics large-eddy simulation informed by laboratory experiment. *Atmospheric Chemistry and Physics*, *24*(18), 10245-10260. doi: 10.5194/acp-24-10245-2024.
Westbrook, C.D., Hogan, R.J. & Illingworth, A.J. (2008), The capacitance of pristine ice crystals and aggregate snowflakes. *Journal of the atmospheric sciences*, *65*(1), 206-219. doi: 10.1175/2007JAS2315.1.
Yang, F., Ovchinnikov, M. & Shaw, R.A. (2015), Long-lifetime ice particles in mixed-phase stratiform clouds: Quasi-steady and recycled growth. *Journal of Geophysical Research: Atmospheres*, *120*(22), 11-617. doi: 10.1002/2015JD023679.

[Geophysical Research Letters]

Supporting Information for

[Direct Numerical Simulations of Mixed-Phase Cloud Glaciation Caused by Turbulent Mixing]

[Abdullah Al Muti Sharfuddin[1], Foluso Ladeinde[2], Yangang Liu[3], Fan Yang[4], Mohammad Atif[5], Satoshi Endo[6], and Meifeng Lin[7]]

[[1,2]Department of Mechanical Engineering, 113 Light Engineering Building, Stony Brook University, 100 Nicolls Road, Stony Brook, New York, 11794-2300, USA.

[3,4,6]Environmental Science and Technologies Department, Brookhaven National Laboratory, Rutherford Dr, Upton, New York, 11973-5000, USA.

[5,7]Computing and Data Sciences Directorate, Brookhaven National Laboratory, 2 Center St, Upton, New York, 11973-5000, USA.]

Contents of this file

Text S1 and Table S1.

Text S1.

We derive the expression for temporal evolution of the ice mass fraction in a mixed-phase cloud. Let the total mass of ice crystals be M_{ice} and that of cloud droplets be M_w. The total mixed-phase cloud mass M_{total} is:

$$M_{total} = M_w + M_{ice}\,, \tag{S1}$$

The ice mass fraction χ_{ice} is:

$$\chi_{ice} = \frac{M_{ice}}{M_{total}}. \tag{S2}$$

The growth equations are rewritten as:

$$\frac{dM_{ice}}{dt} = 4\pi\rho_{ice}\sum_{i=1}^{n_{ice}}\Big(G_{ice}r_{ice}(t)S_{e,ice}(\boldsymbol{X},t)\Big)_i\,, \tag{S3}$$

$$\frac{dM_w}{dt} = 4\pi\rho_w\sum_{i=1}^{n_w}\Big(G_w r_w(t)S_{e,w}(\boldsymbol{X},t)\Big)_i\,. \tag{S4}$$

Taking the time derivative of χ_{ice} using the quotient rule:

$$\frac{d\chi_{ice}}{dt} = \frac{1}{M_{total}}\frac{dM_{ice}}{dt} - \frac{M_{ice}}{M_{total}^2}\frac{dM_{total}}{dt}\,, \tag{S5}$$

Also,

$$\frac{dM_{total}}{dt} = \frac{dM_{ice}}{dt} + \frac{dM_w}{dt}\,, \tag{S6}$$

Then,

$$\frac{d\chi_{ice}}{dt} = \frac{1}{M_{total}}\left(\frac{dM_{ice}}{dt} - \frac{M_{ice}}{M_{total}}\left(\frac{dM_{ice}}{dt} + \frac{dM_w}{dt}\right)\right),$$

$$\frac{d\chi_{ice}}{dt} = \frac{1}{M_{total}}\left(\dot{M}_{ice} - \chi_{ice}\left(\dot{M}_{ice} + \dot{M}_w\right)\right) = \frac{1}{M_{total}}\left((1-\chi_{ice})\dot{M}_{ice} - \chi_{ice}\dot{M}_w\right), \tag{S7}$$

Then,

$$\frac{d\chi_{ice}}{dt} = \frac{4\pi}{M_{total}}\left((1-\chi_{ice})\rho_{ice}\sum_{i=1}^{n_{ice}}\left(G_{ice}r_{ice}S_{e,ice}\right)_i - \chi_{ice}\rho_w\sum_{i=1}^{n_w}\left(G_w r_w S_{e,w}\right)_i\right). \tag{S8}$$

Equation (S8) shows that χ_{ice} increases if $S_{e,ice}$ is positive but $S_{e,w}$ is negative at the same time. Next, we derive the expression for χ_{ice}. The bulk growth rates can be defined as:

$$\dot{M}_{ice} = k_{ice}M_{ice} = k_{ice}\chi_{ice}M_{total}\,, \tag{S9}$$

$$\dot{M}_w = -k_w M_w = -k_w(1-\chi_{ice})M_{total}\,. \tag{S10}$$

Then Equation (S7) becomes,

$$\frac{d\chi_{ice}}{dt} = \frac{1}{M_{total}}\left((1-\chi_{ice})k_{ice}\chi_{ice}M_{total} - \chi_{ice}(-k_w)(1-\chi_{ice})M_{total}\right),$$

$$\frac{d\chi_{ice}}{dt} = k_{ice}\chi_{ice}(1-\chi_{ice}) + k_w\chi_{ice}(1-\chi_{ice}),$$

$$\frac{d\chi_{ice}}{dt} = (k_{ice}+k_w)\chi_{ice}(1-\chi_{ice})\,. \tag{S11}$$

The logistic differential equation is:

$$\frac{d\chi_{ice}}{dt} = K\chi_{ice}(1-\chi_{ice}), \quad K = k_{ice}+k_w\,, \tag{S12}$$

Integrating Equation (S12) with respect to time yields the Sigmoid curve:

$$\chi_{ice}(\mathrm{t}) = \frac{1}{1+\left(\frac{1-\chi_{ice,0}}{\chi_{ice,0}}\right)e^{-Kt}} = \frac{1}{1+e^{-K(t-t_0)}}\,. \tag{S13}$$

Above, t_0 is the time at which the mixed-phase cloud is exactly half-glaciated ($\chi_{ice} = 0.5$). Based on Equation (S13), the evolution profile of χ_{ice} should follow the 'S' shape of a Sigmoid curve. We fit the parameters y_a, K, and t_0 for the five cases in Figure 1(d) and complete the following table S1. Note that y_a is the asymptotic value obtained from Figure 1(d).

Table S1.

Parameter fits for the Sigmoid function

Case	y_a	$K\ (1/s)$	$t_0\ (s)$	The function
H-S-H	1.00	0.30	36.0	$\frac{1}{1+e^{-0.30(t-36)}}$
H-S-L	0.98	0.16	45.0	$\frac{1}{1+e^{-0.16(t-45)}}$
L-C-L	0.93	0.19	52.0	$\frac{1}{1+e^{-0.19(t-52)}}$

L-P-L	0.92	0.18	52.5	$\frac{1}{1+e^{-0.18(t-52.5)}}$
L-S-L	0.91	0.18	53.0	$\frac{1}{1+e^{-0.18(t-53)}}$